<u>Example</u> ESR = electron spin resonance

$$H = \underbrace{\sum_{ij} J_{ij}\, \vec{S}_i \cdot \vec{S}_j}_{H_0,\ \text{hamiltonian of interacting spins}} - \underbrace{\sum_i \vec{B} \cdot \vec{S}_i}_{\text{external static magnetic field}} - \underbrace{\sum_i \vec{h}_i(t) \cdot \vec{S}_i}_{H',\ \text{time-dependent perturbation}}$$

Usually, $\vec{B}$ and $\vec{h}_i$ are uniform fields (wave length $\to \infty$)

$\Rightarrow$ can re-write H in terms of the total spin $\vec{M}$

$$H = \underbrace{H_0 - \vec{B} \cdot \vec{M}}_{H} - \vec{h}(t) \cdot \vec{M} \quad ; \quad \vec{M} \equiv \sum_i \vec{S}_i$$

let $\vec{B} = B\hat{z}$ and $\vec{h}(t) = h_0(\cos\omega t\, \hat{x} - \sin\omega t\, \hat{y})$ circularly polarized EM wave

$$H'(t) = \frac{1}{2} h_0 \left(e^{i\omega t} + e^{-i\omega t}\right) \hat{M}^x - \frac{1}{2i} h_0 \left(e^{i\omega t} - e^{-i\omega t}\right) \hat{M}^y =$$

$$= \frac{1}{2} h_0 \left\{ e^{i\omega t} \left(\hat{M}^x + i\hat{M}^y\right) + e^{-i\omega t} \left(\hat{M}^x - i\hat{M}^y\right) \right\}$$

$$= \frac{1}{2} h_0 \left\{ e^{i\omega t} \hat{M}^+ + e^{-i\omega t} \hat{M}^- \right\}.$$

Thus here $h = h^* = \frac{1}{2} h_0$, $\hat{O} = \hat{M}^+$, $\hat{O}^+ = \hat{M}^-$

$$\overline{\frac{dE}{dt}} = -2\omega \left|\frac{h_0}{2}\right|^2 \operatorname{Im}\left\{ \chi_{M^+M^-}(\omega) \right\}$$

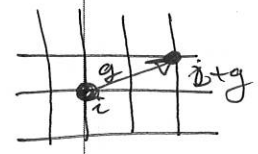
$$\chi_{M^+M^-}(t) = -\frac{i}{\hbar}\, \theta(t) \left\langle \left[\hat{M}^+(t), \hat{M}^-(0) \right] \right\rangle$$

$$\hat{M}^+(t) = e^{iHt/\hbar}\, \hat{M}^+\, e^{-iHt/\hbar}$$

$\hat{H} \equiv \hat{H}_0 - B\hat{M}^z$ (total t-independent Hamiltonian)

$$\frac{d}{dt} \hat{M}^+(t) = \frac{i}{\hbar} \left[\hat{H}, \hat{M}^+ \right]$$

Consider now $H_0 = \frac{1}{2}\sum_{i,g} J_g \vec{S}_i \cdot \vec{S}_{i+g} =$



$$= \frac{1}{2}\sum_{i,g} J_g \left(S_i^z S_{i+g}^z + \frac{1}{2}\left(S_i^+ S_{i+g}^- + S_i^- S_{i+g}^+\right)\right)$$

$$\hat{M}^+ = \sum_i S_i^+$$

Commutation relations $[S_i^z, S_j^+] = \delta_{ij} S_j^+$

$$[S_i^z, S_j^-] = -\delta_{ij} S_j^-$$

$$[S_i^+, S_j^-] = 2 S_i^z \delta_{ij}$$

$$\Rightarrow [\hat{H}_0, \hat{M}^+] = \frac{1}{2}\sum_{ig} J_g \sum_n \Big([S_i^z S_{i+g}^z, S_n^+]$$

$$+ \frac{1}{2}[S_i^+ S_{i+g}^-, S_n^+] + \frac{1}{2}[S_i^- S_{i+g}^+, S_n^+] \Big)$$

$$S_i^z \underbrace{S_{i+g}^z S_n^+}_{S_n^+ \delta_{n,i+g} + S_n^+ S_{i+g}^z} - S_n^+ S_i^z S_{i+g}^z = S_i^z S_n^+ \delta_{n,i+g}$$

$$+ \underbrace{S_i^z S_n^+}_{\delta_{ni} S_n^+ + S_n^+ S_{i+g}^z} S_{i+g}^z - S_n^+ S_i^z S_{i+g}^z = S_i^z S_n^+ \delta_{n,i+g} + S_n^+ S_{i+g}^z \delta_{n,i}$$

$$[S_i^+ S_{i+g}^-, S_n^+] = S_i^+ S_{i+g}^- S_n^+ - \underbrace{S_n^+ S_i^+ S_{i+g}^-}_{S_i^+ \underbrace{S_n^+ S_{i+g}^-}_{2 S_n^z \delta_{n,i+g} + S_{i+g}^- S_n^+}} =$$

$$= -2 S_i^+ S_n^z \delta_{n,i+g}$$

$$[S_i^- S_{i+g}^+, S_n^+] = -2 S_n^z S_{i+g}^+ \delta_{n,i}$$

As a result $[\hat{H}_0, \hat{M}^+] = \frac{1}{2} \sum_{ig} \sum_n J_g \Big\{ \underline{S_i^z S_n^+} \delta_{n,i+g}$

$+ \underline{\underline{S_n^+ S_{i+g}^z}} \delta_{n,i} + \frac{1}{2} \Big(-2 \underline{\underline{S_i^+ S_n^z}} \delta_{n,i+g} - 2 \underline{S_n^z S_{i+g}^+} \delta_{n,i} \Big) \Big\}$

$= 0$

$\hat{M}^+$ commutes with $\hat{H}_0$ because it is $\underline{SU(2) \text{ symmetric}}$
(spin rotation invariant)

Remains to check $[-B \hat{M}^z, \hat{M}^+] = -B \sum_{n,m} \underbrace{[S_m^z, S_n^+]}_{S_n^+ \delta_{nm}} =$

$$= -B \sum_n S_n^+ = -B \hat{M}^+$$

Thus we obtain a very simple result

$$\frac{d}{dt} \hat{M}^+(t) = \frac{i}{\hbar} [\hat{H}, \hat{M}^+] = -\frac{i}{\hbar} B \hat{M}^+(t)$$

$$\Rightarrow \hat{M}^+(t) = e^{-iBt/\hbar} \hat{M}^+(0)$$

$$\Rightarrow \chi_{M^+M^-}(t) = -\frac{i}{\hbar} \theta(t) \langle [e^{-iBt/\hbar} \hat{M}^+(0), \hat{M}^-(0)] \rangle =$$

$$= -\frac{i}{\hbar} \theta(t) e^{-iBt/\hbar} \langle [\hat{M}^+, \hat{M}^-] \rangle$$

But $[M^+, M^-] = \sum_{n,m} [S_n^+, S_m^-] = \sum_{n,m} 2 S_n^z \delta_{n,m} =$

$$= 2 \sum_n S_n^z = 2 \hat{M}^z.$$

Therefore $\chi_{M^+M^-}(\omega) = \int_{-\infty}^{\infty} dt \, e^{i\omega t - \delta t} \chi_{M^+M^-}(t) =$

$$= -\frac{i}{\hbar} \int_0^{\infty} dt \; 2 \langle \hat{M}^z \rangle e^{i\omega t - iBt/\hbar} e^{-\delta t} =$$

$$= -\frac{2i}{\hbar} \langle \hat{M}^z \rangle \int_0^{\infty} dt \; e^{it(\omega + i\delta - \frac{B}{\hbar})}$$

$$\chi_{M^+M^-}(\omega) = -\frac{2i}{\hbar}\langle \hat{M}^z\rangle \left.\frac{e^{it(\omega+i\delta-B/\hbar)}}{i(\omega+i\delta-\frac{B}{\hbar})}\right|_0^\infty =$$

$$= \frac{2\langle M^z\rangle}{\hbar\left(\omega+i\delta-\frac{B}{\hbar}\right)} = \frac{2\langle M^z\rangle}{\hbar}\left\{P\frac{1}{\omega-B/\hbar} - i\pi\,\delta(\omega-B/\hbar)\right\}$$

$$\Rightarrow \operatorname{Im}\chi_{M^+M^-}(\omega) = -\frac{2\pi}{\hbar}\langle M^z\rangle\,\delta(\omega-B/\hbar)$$

Thus we find an <u>exact</u> result (which is very rare) which is specific to spin-rotation invariant systems ~~Hamilto~~

$$\overline{\frac{dE}{dt}} = -2\omega\left|\frac{h_0}{2}\right|^2 \frac{-2\pi}{\hbar}\langle\hat{M}^z\rangle\,\delta\left(\omega-\frac{B}{\hbar}\right) =$$

$$= \frac{\pi h_0^2}{\hbar}\underbrace{\langle\hat{M}^z\rangle}_{\substack{\text{equilibrium}\\ \text{magnetization}\\ \text{(function of } B \text{ and } T)}}\,\delta(\omega-B/\hbar)$$

The energy is absorbed at the Zeeman frequency $\omega_Z = \frac{1}{\hbar}B$ which "knows" nothing about interactions J_g in the system!

(we get the same result for a single paramagnetic spin!)

This is true for every (a) $SU(2)$ invariant H_0;

(b) also assumed is paramagnetic (disordered) state

$\underline{T>T_c}$

i.e. with no spontaneously broken $SU(2)$ symmetry

(c) uniform B, which means $\vec{q}=0$ response.

Not so if H contains spin anisotropies $\Rightarrow$ ESR probes anisotropies!